

Sean Luka Girgis

Senior Performance Engineer

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Professional Summary

Senior Performance Engineer with 20+ years of enterprise IT experience across banking, financial services, telecom, hospitality, and large-scale production environments. Strong background in performance testing strategy, LoadRunner/Performance Center, JMeter-style testing, Java/J2EE diagnostics, thread and heap analysis, garbage collection analysis, Dynatrace/APM monitoring, Python scripting, SQL/Oracle, and stakeholder-ready performance reporting. Experienced defining non-functional requirements, building performance test plans, analyzing bottlenecks across application/database/infrastructure layers, and translating findings into actionable optimization recommendations.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built automated Python/Pandas telemetry pipelines ingesting P95 performance and capacity data from 6,000+ endpoints, replacing manual processes with repeatable performance analytics and reporting workflows.
- Analyzed infrastructure, application, and capacity metrics to identify bottlenecks, utilization risk, peak patterns, and optimization opportunities across enterprise banking systems.
- Integrated BMC TrueSight/TSCO, CA APM, and AppDynamics data into unified Oracle reporting datasets for capacity, performance, and executive-level operational visibility.
- Designed optimized Oracle schemas and SQL reporting structures to support historical retention, performance trend analysis, peak forecasting, and operational risk review.
- Developed Prophet and scikit-learn forecasting models to predict capacity bottlenecks months ahead and support proactive remediation planning.
- Produced dashboards and reports using Streamlit, matplotlib, seaborn, and plotly to communicate performance trends, capacity risk, and recommendations to senior stakeholders.
- Supported root-cause analysis by correlating monitoring data, workload behavior, application performance signals, and infrastructure KPIs.
- Documented repeatable performance reporting processes, validation rules, and operational runbooks to improve governance and reduce manual effort.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, supporting production performance monitoring and engineering analysis.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and provide actionable recommendations to engineering teams.
- Integrated HP Performance Center with Dynatrace to correlate load-test behavior, APM traces, infrastructure metrics, and user-experience indicators.
- Implemented End-User Monitoring and end-to-end transaction tracing across web and mobile systems to improve visibility into user journeys and application performance.

- Created performance reports and dashboards that translated telemetry into findings for engineering, infrastructure, and stakeholder review.
- Participated in AWS cloud migration readiness discussions by evaluating monitoring coverage, operational risks, and performance data requirements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM subject matter expert for a large financial-services enterprise with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed dashboards, alerts, SLA thresholds, and management modules used to monitor enterprise application performance and support operational decision-making.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and reduce manual weekly exports.
- Partnered with J2EE/WebLogic and infrastructure teams to diagnose bottlenecks using APM telemetry, JVM indicators, SQL/Oracle signals, thread behavior, and infrastructure metrics.
- Supported large-scale CA APM upgrades and monitoring standardization while maintaining production visibility and minimizing disruption.
- Delivered technical documentation, onboarding guidance, and client training for performance-monitoring standards, agent deployments, and troubleshooting practices.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom web applications under load to identify throughput limits, response-time degradation, and resource bottlenecks including JDBC connections, threads, heap memory, CPU, and garbage collection.
- Used LoadRunner/Performance Center-style performance testing practices to support load, stress, and regression performance validation for enterprise applications.
- Installed and configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to provide real-time visibility during test execution.
- Documented performance baselines, regression thresholds, findings, and recommendations used by engineering teams for release-gate decisions.
- Collaborated with application, testing, and infrastructure teams to troubleshoot performance defects and validate remediation outcomes.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining strict performance and reliability expectations.
- Optimized core transaction processing using C++, OCCI, OCI, SQL, and Oracle, reducing physical hardware footprint while preserving sub-second query latency.
- Built C++/OCCI/OCI validation tooling to verify migration correctness and performance-sensitive behavior across large transaction datasets.
- Performed SQL and database-performance analysis to support high-volume transaction processing and migration readiness.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed high-availability telecom billing interfaces using C, C++, Pro*C, PL/SQL, Oracle, XML, socket programming, POSIX threads, and Unix environments.

- Improved throughput and database performance in billing processes through SQL tuning, sequence optimization, memory reduction, and interface-level performance analysis.
- Built and maintained integration components using enterprise messaging and interface patterns, including MQ Series and XML-based workflows.
- Automated WebLogic/WebSphere administration and deployment support with Korn Shell scripting, reducing manual operational effort.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Flagship Projects

Performance Telemetry and Capacity Reporting Platform

Built Python and SQL-driven telemetry reporting workflows to transform enterprise infrastructure and application performance data into dashboards, forecasts, and executive-ready findings.

Technologies: Python, SQL, Oracle, BMC TrueSight, AppDynamics, CA APM, Streamlit

FAST Project

Mined real-user and synthetic monitoring data from Dynatrace and Gomez to identify performance-improvement opportunities for critical revenue-generating systems.

Technologies: Dynatrace, Gomez Synthetic Monitoring, Performance Analytics, Telemetry Reporting

J2EE Performance Diagnostics and Release Validation

Supported load and regression performance validation for J2EE applications using LoadRunner/Performance Center-style practices, JMX monitoring, Wily Introscope, and thread-dump analysis.

Technologies: LoadRunner, Performance Center, J2EE, JMX, Wily Introscope, Thread Dumps

High-Volume Database Migration Validation

Built validation and performance-sensitive migration tooling for a high-throughput travel shopping engine moving from MySQL nodes to Oracle RAC.

Technologies: C++, OCCI, OCI, Oracle RAC, SQL, Performance Validation